Noise Effect on the Flux Calibration Parameter $\hat{\ell}$

Step 1: Expansion of J about $\hat{\ell}_0$

Perturbation. Shift the length scale by a small amount $\delta\hat{\ell}$, i.e. $\hat{\ell} = \hat{\ell}_0 + \delta\hat{\ell}$, and expand J in a Taylor series about $\hat{\ell}_0$:

$$J(\hat{\ell}_0 + \delta\hat{\ell}) = J(\hat{\ell}_0) + J'(\hat{\ell}_0)\,\delta\hat{\ell} + \tfrac{1}{2}\,J''(\hat{\ell}_0)\,\delta\hat{\ell}^{\,2} + O\Big(\delta\hat{\ell}^{\,3}\Big)$$

First-order term. Since $\hat{\ell}_0$ is a minimizer of J,

$$J'(\hat{\ell}_0) = 0$$

Result. The increase of the cost caused by the shift $\delta\hat{\ell}$ is governed by the curvature $J''(\hat{\ell}_0)$:

$$\Delta J \equiv J(\hat{\ell}_0 + \delta\hat{\ell}) - J(\hat{\ell}_0) \approx \tfrac{1}{2}\,J''(\hat{\ell}_0)\,\delta\hat{\ell}^{\,2}$$

Step 2: Residual form of J

$$\mathbf{W} = \sum_{i=1}^{N_c} \mathbf{S}_i^T \mathbf{S}_i, \qquad \mathbf{c}_j = \sum_{i=1}^{N_c} \mathbf{S}_i^T \mathbf{b}_{ij}, \qquad \mathbf{g}_j = \mathbf{W}^{-1}\mathbf{c}_j$$

$$\boldsymbol{\varepsilon}_{ij} = \mathbf{b}_{ij} - \mathbf{S}_i\,\mathbf{g}_j$$

$$\sum_{i=1}^{N_c} \boldsymbol{\varepsilon}_{ij}^T \boldsymbol{\varepsilon}_{ij} = \sum_{i=1}^{N_c} \mathbf{b}_{ij}^T \mathbf{b}_{ij} - \mathbf{c}_j^T \mathbf{W}^{-1} \mathbf{c}_j$$

$$J(\hat{\ell}) = \sum_{j=1}^{N_I} J_j, \qquad J_j = \sum_{i=1}^{N_c} \boldsymbol{\varepsilon}_{ij}^T \boldsymbol{\varepsilon}_{ij}$$

Excitations are independent; within one excitation the points are coupled through $\mathbf{g}_j$.

Step 3: First derivative of J_j

A prime denotes $d/d\hat{\ell}$.

$$J_j' = \sum_{i=1}^{N_c} \frac{\partial J_j}{\partial \boldsymbol{\varepsilon}_{ij}}\,\boldsymbol{\varepsilon}_{ij}' = 2\sum_{i=1}^{N_c} \boldsymbol{\varepsilon}_{ij}^T \boldsymbol{\varepsilon}_{ij}'$$

$$\boldsymbol{\varepsilon}_{ij}' = -\mathbf{S}_i'\,\mathbf{g}_j - \mathbf{S}_i\,\mathbf{g}_j'$$

$$\mathbf{d}_{ik} = \bar{\mathbf{p}}_i - \bar{\mathbf{p}}_{ck}, \qquad s_{ik,x}' = -\frac{1}{\hat{\ell}}\left[\frac{\bar{x}_i}{\|\mathbf{d}_{ik}\|^3} - \frac{3\,(\bar{x}_i - \bar{x}_{ck})\,\mathbf{d}_{ik}^T \bar{\mathbf{p}}_i}{\|\mathbf{d}_{ik}\|^5}\right] \quad (y,\, z \text{ similarly})$$

$$\mathbf{s}_{ik}' = -\frac{1}{\hat{\ell}\,\|\mathbf{d}_{ik}\|^3}\left(\mathbf{E}_3 - 3\,\frac{\mathbf{d}_{ik}\mathbf{d}_{ik}^T}{\|\mathbf{d}_{ik}\|^2}\right)\bar{\mathbf{p}}_i, \qquad \mathbf{S}_i' = \begin{bmatrix}\mathbf{s}_{i1}' & \cdots & \mathbf{s}_{i6}'\end{bmatrix}$$

where $\mathbf{E}_3$ is the 3×3 identity matrix. From $\mathbf{W}\mathbf{g}_j = \mathbf{c}_j$:

$$\mathbf{g}_j' = \begin{bmatrix} g_{j1}' & \cdots & g_{j6}' \end{bmatrix}^T = \mathbf{W}^{-1} \sum_{i=1}^{N_c} \left(\mathbf{S}_i'^T \boldsymbol{\varepsilon}_{ij} - \mathbf{S}_i^T \mathbf{S}_i'\,\mathbf{g}_j\right)$$

Since $\sum_i \mathbf{S}_i^T \boldsymbol{\varepsilon}_{ij} = \mathbf{c}_j - \mathbf{W}\mathbf{g}_j = \mathbf{0}$:

$$J_j' = -2\sum_{i=1}^{N_c} \boldsymbol{\varepsilon}_{ij}^T\,\mathbf{S}_i'\,\mathbf{g}_j$$

Step 4: Second derivative of J_j

$$J_j'' = 2\sum_{i=1}^{N_c}\left(\boldsymbol{\varepsilon}_{ij}'^T\boldsymbol{\varepsilon}_{ij}' + \boldsymbol{\varepsilon}_{ij}^T\boldsymbol{\varepsilon}_{ij}''\right)$$

$$\boldsymbol{\varepsilon}_{ij}'' = -\left(\mathbf{S}_i''\,\mathbf{g}_j + 2\,\mathbf{S}_i'\,\mathbf{g}_j' + \mathbf{S}_i\,\mathbf{g}_j''\right), \qquad \mathbf{S}_i'' = \begin{bmatrix}\mathbf{s}_{i1}'' & \cdots & \mathbf{s}_{i6}''\end{bmatrix}, \qquad \mathbf{g}_j'' = \begin{bmatrix}g_{j1}'' & \cdots & g_{j6}''\end{bmatrix}^T$$

where $\mathbf{s}_{ik}''$ is obtained by symbolic differentiation of $\mathbf{s}_{ik}$ with respect to $\hat{\ell}$. Substituting $\boldsymbol{\varepsilon}_{ij}''$:

$$\sum_{i=1}^{N_c}\boldsymbol{\varepsilon}_{ij}^T\boldsymbol{\varepsilon}_{ij}'' = -\sum_{i=1}^{N_c}\boldsymbol{\varepsilon}_{ij}^T\left(\mathbf{S}_i''\,\mathbf{g}_j + 2\,\mathbf{S}_i'\,\mathbf{g}_j'\right) - \Big(\sum_{i=1}^{N_c}\mathbf{S}_i^T\boldsymbol{\varepsilon}_{ij}\Big)^T\mathbf{g}_j''$$

Since $\sum_i \mathbf{S}_i^T\boldsymbol{\varepsilon}_{ij} = \mathbf{0}$, the last term vanishes:

$$J_j'' = 2\sum_{i=1}^{N_c}\|\boldsymbol{\varepsilon}_{ij}'\|^2 - 2\sum_{i=1}^{N_c}\boldsymbol{\varepsilon}_{ij}^T\left(\mathbf{S}_i''\,\mathbf{g}_j + 2\,\mathbf{S}_i'\,\mathbf{g}_j'\right), \qquad J'' = \sum_{j=1}^{N_I}J_j''$$

Step 5: Noise

$$\tilde{\mathbf{b}}_{ij} = \mathbf{b}_{ij} + \mathbf{n}_{ij}$$

Let $\tilde{J}$ be J with $\mathbf{b}_{ij}$ replaced by $\tilde{\mathbf{b}}_{ij}$; its minimizer is $\hat{\ell}_0 + \delta\hat{\ell}$. Expanding about $\hat{\ell}_0$ ($\tilde{J}'(\hat{\ell}_0) \neq 0$ because of the noise):

$$\tilde{J}(\hat{\ell}_0 + \delta\hat{\ell}) = \tilde{J}(\hat{\ell}_0) + \tilde{J}'(\hat{\ell}_0)\,\delta\hat{\ell} + \tfrac{1}{2}\,\tilde{J}''(\hat{\ell}_0)\,\delta\hat{\ell}^2 + O\Big(\delta\hat{\ell}^3\Big)$$

Minimizing over $\delta\hat{\ell}$:

$$\frac{\partial\tilde{J}}{\partial\,\delta\hat{\ell}} = \tilde{J}'(\hat{\ell}_0) + \tilde{J}''(\hat{\ell}_0)\,\delta\hat{\ell} + O\Big(\delta\hat{\ell}^2\Big) = 0 \;\Rightarrow\; \delta\hat{\ell} \approx -\frac{\tilde{J}'(\hat{\ell}_0)}{\tilde{J}''(\hat{\ell}_0)}$$

where, evaluated at $\hat{\ell}_0$,

$$\tilde{J}' = -2\sum_{j=1}^{N_I}\sum_{i=1}^{N_c}\tilde{\boldsymbol{\varepsilon}}_{ij}^T\,\mathbf{S}_i'\,\tilde{\mathbf{g}}_j$$

$$\tilde{J}'' = \sum_{j=1}^{N_I}\left[2\sum_{i=1}^{N_c}\|\tilde{\boldsymbol{\varepsilon}}_{ij}'\|^2 - 2\sum_{i=1}^{N_c}\tilde{\boldsymbol{\varepsilon}}_{ij}^T\left(\mathbf{S}_i''\,\tilde{\mathbf{g}}_j + 2\,\mathbf{S}_i'\,\tilde{\mathbf{g}}_j'\right)\right]$$

Result

Linearized in the noise, all quantities evaluated at $\hat{\ell}_0$ with the noise-free $\mathbf{b}_{ij}$:

$$\delta\hat{\ell} \approx -\frac{\tilde{J}'(\hat{\ell}_0)}{\tilde{J}''(\hat{\ell}_0)} \approx -\frac{2}{J''(\hat{\ell}_0)}\sum_{j=1}^{N_I}\sum_{i=1}^{N_c}\boldsymbol{\varepsilon}_{ij}'^T\mathbf{n}_{ij} = \boldsymbol{\mathcal{E}}^T\mathbf{N}$$

$$\boldsymbol{\mathcal{E}} = -\frac{2}{J''(\hat{\ell}_0)}\begin{bmatrix}\boldsymbol{\varepsilon}_{11}'\\ \boldsymbol{\varepsilon}_{21}'\\ \vdots\\ \boldsymbol{\varepsilon}_{N_cN_I}'\end{bmatrix}, \qquad \mathbf{N} = \begin{bmatrix}\mathbf{n}_{11}\\ \mathbf{n}_{21}\\ \vdots\\ \mathbf{n}_{N_cN_I}\end{bmatrix} \qquad (3N_cN_I \times 1)$$

With $E[\mathbf{n}_{ij}] = \mathbf{0}$, independent $\mathbf{n}_{ij}$, and $\mathrm{Cov}[\mathbf{n}_{ij}] = \sigma^2\mathbf{E}_3$:

$$\sigma_{\hat{\ell}} = \sigma_{\delta\hat{\ell}} = \frac{2\sqrt{\sum_{j=1}^{N_I}\sum_{i=1}^{N_c}\|\boldsymbol{\varepsilon}_{ij}'\|^2}}{J''(\hat{\ell}_0)}\,\sigma$$